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RAPID: Robust APT detection and investigation using context-aware deep learning

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Abstract

Advanced persistent threats (APTs) pose a critical cybersecurity challenge, enabling attackers to maintain long-term unauthorized access while evading detection. Current APT detection approaches struggle with three key limitations: high false positive rates that lead to alert fatigue, poor adaptability to evolving system behaviors, and the inability to provide actionable investigation context. We present RAPID, a novel deep learning framework that addresses these challenges through context-aware anomaly detection and intelligent alert tracing. RAPID's key innovation lies in its dual-phase architecture: first, it employs self-supervised sequence learning with iteratively updated embeddings to capture dynamic system behavior patterns; second, it leverages these embeddings to reconstruct precise attack narratives through provenance graph analysis. Our comprehensive evaluation across five diverse real-world datasets demonstrates RAPID's effectiveness, achieving up to 74% precision with near-perfect recall while using only 30% of the data for training, substantially outperforming state-of-the-art methods that require 80% training data to achieve similar performance levels. The framework automatically generates detailed attack narratives that enable efficient incident response, significantly outperforming existing approaches in both detection accuracy and alert investigation precision.

Introduction

Advanced persistent threats (APTs) are highly sophisticated and targeted cyberattacks that pose significant risks to organizations, leading to data breaches, intellectual property theft, financial loss, and reputational damage. Recent high-profile incidents, such as the exploitation of critical vulnerabilities in Microsoft Exchange Servers [1] and the 3CX supply chain attack [2], highlight the persistent evolution of APT techniques. These events underscore the urgent need for robust and adaptive APT detection and investigation capabilities to counteract the growing complexity of these threats.

Provenance data analysis has emerged as a promising approach for APT detection and investigation. Rule-based methods for provenance data analysis [3], [4], [5], [6] can detect known attack patterns but are impractical due to the need for expert labeling and their inability to detect zero-day attacks. To address these limitations, many studies have investigated anomaly-based methods [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21]. These methods, which focus on learning normal system behavior and detecting deviations, are capable of identifying novel attack vectors without the need for attack data, which may not always be available.

However, in the practical application of anomaly-based methods alert fatigue remains a primary challenge, as it can be difficult to maintain an acceptable balance between high recall and a low false positive rate due to their sensitivity to normal variability (natural noise) in system behavior. While several recent studies [7], [16], [17], [18], [19], [21], [22] proposing anomaly-based methods for APT detection have reported good results, the proposed methods have two main limitations: (1) they rely on large training datasets and disproportionately small testing windows, which does not align with the dynamic and continuously evolving nature of real-world data environments, and (2) they suffer from data leakage [17], [19], since the models were inadvertently trained on future data, which may result in exaggerated detection performance. Thus, although these methods produced good results, further research is needed to address their limitations and improve their effectiveness and reliability; addressing these issues will increase their adoption and real-world application.

In light of the above-mentioned research gaps and limitations, we propose RAPID, a robust anomaly-based APT detection and investigation framework that leverages context-aware deep learning on provenance data to reduce alert fatigue and ensure high recall in evolving system environments. RAPID combines self-supervised learning and provenance analysis to account for the dynamic nature of evolving systems, offering a practical and effective solution for real-world application. This integration enables RAPID to detect unknown attacks with very low false alarms under realistic conditions (i.e., ensuring no data leakage), thus addressing critical gaps in existing APT detection methods.

RAPID implements a modular architecture consisting of two main phases: detection and tracing. The detection phase first constructs dense vector representations (embeddings) of system entities based on their interaction patterns. These embeddings capture the semantic relationships between processes, files, and network connections, providing a rich contextual foundation for anomaly detection. By decoupling entity representation learning from temporal analysis, RAPID can

efficiently update these embeddings during inference without retraining the entire model, enabling continuous adaptation to changing system behaviors.

The sequence analysis component then utilizes these embeddings to enhance its anomaly detection capabilities, analyzing temporal patterns within the enriched event sequences. This modular design offers dual benefits: it improves detection accuracy by combining rich contextual representations with temporal pattern analysis, while enabling efficient system adaptation through selective updates to entity representations. This approach addresses a fundamental limitation of existing methods, which require costly complete model retraining to adapt to evolving system behavior.

In the tracing phase, RAPID grounds the detected anomalies in the system's provenance graph to reconstruct attack narratives. For each detected anomaly, the system performs bidirectional graph traversal guided by entity embeddings to identify causally related events. This provenance-based analysis, combined with entity similarity in the embedding space, enables RAPID to filter out irrelevant events while preserving the critical elements of the attack chain. The result is a precise, contextualized alert that captures the complete progression of the attack within the system.

To validate RAPID's effectiveness in real-world settings, we conducted an extensive evaluation using five diverse and well-established datasets: the CADETS, THEIA, CLEARSCOPE, OpTC [23], and Public Arena [24] datasets. Unlike previous studies that introduced data leakage and used unrealistic train-test splits, we employ a evaluation strategy that closely emulates real-world scenarios and mitigates data leakage.

Our experiments show that RAPID consistently outperforms state-of-the-art methods across multiple granularity levels, achieving near-perfect detection accuracy, significantly reducing false positives, and generating precise alerts that facilitate efficient incident response and investigation.

The main contributions of this paper are:

- **Modular Learning Architecture:** A novel separation of entity representation learning from temporal sequence analysis that enables efficient model adaptation to evolving system behaviors without requiring complete retraining.
- **Embedding-Guided Alert Reconstruction:** An effective technique for reconstructing attack narratives by combining provenance graph analysis with learned entity representations, resulting in precise attack chain identification with 74% precision at near-perfect recall.
- **Realistic Evaluation Framework:** A methodologically sound evaluation approach using limited training data (30%) and extended testing periods that better reflects real-world deployment conditions, addressing previous methodological limitations in APT detection research.

- **Open-sourced and Reproducible Framework:** We release the full RAPID codebase, including data preparation, training, and evaluation scripts, to facilitate reproducibility and future research.²
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Section snippets

Background

The increasing sophistication of APTs necessitates the development of robust and effective detection methods. Provenance graphs have become an important tool in this domain [25], [26]. This section begins with a motivating example of a provenance graph, which is followed by a description of the assumed threat model, and concludes with a review of prior research, highlighting its limitations.

Motivating Example.Provenance graphs are a common approach for representing system audit data [25], [26]. ...

Prior research

Provenance-based methods for APT detection were developed to counter increasingly complex cybersecurity threats. Rule-based techniques [4], [8] employ predefined security policies and heuristic rules to pinpoint attack patterns. While these methods provide detailed fine-grained event-level detection [26] and may have low false positive rates, they require substantial manual intervention and struggle to detect zero-day exploits [11].

Anomaly-based methods, including statistical approaches [3], [6] ...

Overview

RAPID implements a multi-stage architecture for APT detection and investigation, optimizing the balance between detection accuracy and computational efficiency. The framework processes system audit data through four key stages, as illustrated in Fig. 2.

The workflow begins with provenance graph construction from system audit logs (A), which transforms system-level events into a data transfer layer representation. This abstraction maintains essential causal relationships while reducing storage ...

Evaluation setup

In this section, we provide a detailed description of the experimental setup and method used to obtain the ground truth. RAPID 's performance is evaluated on five diverse public datasets from common auditing systems. We compare its performance to that of state-of-the-art methods analyze the quality and traceability of the alerts generated by RAPID, assess the impact of the object embeddings, and examine its scalability and performance overhead. ...

Attack detection and entity identification

Table 2 presents a comprehensive evaluation of APT detection methods using our consistent 30% training split framework. We analyze performance at both attack window and entity levels to provide a nuanced understanding of detection capabilities.

Attack Window Detection.At the attack window level, RAPID achieves perfect detection (precision=1.00, recall=1.00) across all three datasets. While UNICORN shows similar window-level metrics, its alerts span unnecessarily long time periods, making attack ...

Conclusion

We introduce RAPID, a novel framework that addresses the critical challenge of detecting and investigating advanced persistent threats in modern enterprise environments. The key innovation of RAPID lies in its synergistic integration of context-aware anomaly detection with intelligent alert tracing, leveraging dynamically updated object embeddings to enhance both detection accuracy and attack narrative reconstruction.

Our comprehensive evaluation demonstrates that RAPID significantly outperforms ...

CRedit authorship contribution statement

Yonatan Amaru: Writing – review & editing, Writing – original draft, Software, Methodology, Data curation. **Prasanna N. Wudali:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Data curation. **Yuval Elovici:** Writing – review & editing, Supervision, Resources, Project administration. **Asaf Shabtai:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Conceptualization. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

Yonatan Amaru is a master (M.Sc.) student at the Department of Software and Information Systems Engineering at Ben-Gurion University of the Negev (BGU). His research interests include machine learning and data science in the cybersecurity domain. ...

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